

Econ 101A

Section 9: Midterm Review (Solutions)

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This is a review, not new material. The goal is to see the whole unit as one connected picture and to close out a few problems we did not finish in section. Nothing here is meant to be hard. If you can follow the map below and redo the short problems, you are in good shape. Take a breath, you know more than you think.

1 The Map: One Consumer Problem, Many Faces

Everything this unit fits on one page.

1. Preferences. Rational means *complete* (you can compare any two bundles) and *transitive* (no cycles). A utility function is just a numbering of bundles that respects the ranking. Any strictly increasing transform of u (like e^u or $\ln u$) represents the same preferences and gives the same demands.

2. Utility maximization. Choose the best affordable bundle:

$$\max_{x,y} u(x,y) \quad \text{s.t.} \quad px + qy = M, \quad \mathcal{L} = u + \lambda(M - px - qy).$$

Interior optimum: tangency $u_x/u_y = p/q$, plus the budget. Watch for the traps: perfect substitutes and other corners, and non-convex preferences where the tangency is a minimum (check a corner or the second-order condition).

3. Marshallian demand and indirect utility. Solving gives $x^*(p, q, M)$ and $v(p, q, M) = u(x^*, y^*)$. By the envelope theorem, $\partial v / \partial M = \lambda$ (marginal utility of income) and $\partial v / \partial p = -\lambda x^*$.

4. The dual: expenditure minimization. Same tangency, different constraint:

$$\min px + qy \quad \text{s.t.} \quad u(x, y) \geq \bar{u} \Rightarrow \text{Hicksian } h(p, q, \bar{u}), \quad e(p, q, \bar{u}) = ph_x + qh_y.$$

Shephard: $\partial e / \partial p = h_x$. Duality: $h_x(p, q, \bar{u}) = x^*(p, q, e)$.

5. Slutsky. The total price effect splits into a substitution part (always ≤ 0) and an income part:

$$\frac{\partial x^*}{\partial p} = \left. \frac{\partial h_x}{\partial p} \right|_{\bar{u}=v} - x^* \frac{\partial x^*}{\partial M}.$$

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Normal good: both terms negative, demand slopes down. Giffen: needs an inferior good with an income effect large enough to flip the total positive. Every Giffen good is inferior; not every inferior good is Giffen.

6. Applications. Same tools, relabeled. Labor supply: leisure priced at the wage w , full income $M + wT$, tangency $u_\ell/u_c = w$, reservation wage below which hours are zero. Intertemporal: future consumption priced at $1/(1+r)$, wealth $W = M_0 + M_1/(1+r)$, Euler $U'(c_0) = \frac{1+r}{1+\delta}U'(c_1)$, smoothing when $r = \delta$.

2 Warm-up Problems

Problem 1 (preferences and transforms)

A consumer's preferences are represented by $u(x, y)$. Does $\tilde{u} = e^u$ represent the same preferences? Would maximizing $\tilde{u}$ subject to the same budget change the optimal bundle?

Solution. Yes, same preferences: e^u is a strictly increasing transform, so $e^{u(a)} \geq e^{u(b)} \iff u(a) \geq u(b)$, and it ranks all bundles identically. The optimal bundle is unchanged, because demand depends only on the ranking, not on the utility numbers. (A *decreasing* transform, like $-u$, would reverse the ranking and is not the same preferences.)

Problem 2 (perfect substitutes)

Let $u(x, y) = 2x + y$ with prices $p, q > 0$ and income M . Find the optimal bundle and say when it is not unique.

Solution. MRS magnitude $= u_x/u_y = 2$, constant, so compare it to p/q . If $2 > p/q$ (i.e. $p < 2q$), spend all on x : $x^* = M/p$, $y^* = 0$. If $2 < p/q$, spend all on y : $y^* = M/q$, $x^* = 0$. At $p/q = 2$ the indifference lines are parallel to the budget line and any bundle on it is optimal.

Problem 3 (Cobb–Douglas quick pass)

Let $u(x, y) = \sqrt{xy}$, prices p, q , income M .

- (a) State the Marshallian demands.
- (b) Is x normal? Is it a gross substitute or complement for y ?

Solution. (a) Equal budget shares: $x^* = \frac{M}{2p}$, $y^* = \frac{M}{2q}$.

(b) $\partial x^*/\partial M = 1/(2p) > 0$, so x is normal. $\partial x^*/\partial q = 0$, so x is neither a gross substitute nor a gross complement for y : with Cobb–Douglas the other price does not enter own demand.

Problem 4 (Giffen versus inferior)

Using the Slutsky equation, explain why a Giffen good must be inferior, and why an inferior good need not be Giffen.

Solution. Slutsky: $\partial x^*/\partial p = (\partial h_x/\partial p) - x^* \partial x^*/\partial M$. The substitution term is always ≤ 0 . Giffen means $\partial x^*/\partial p > 0$, so the income term $-x^* \partial x^*/\partial M$ must be positive and dominant, which needs $\partial x^*/\partial M < 0$: inferior. Conversely, an inferior good with a small income effect keeps the total negative, so it slopes down normally. Inferiority is necessary, not sufficient.

Problem 5 (numerical Slutsky)

Keep $u = \sqrt{xy}$ at $p = q = 1$, $M = 8$, where $x^* = M/2p$, $y^* = M/2q$, and the Hicksian is $h_x = \bar{u}\sqrt{q/p}$. Decompose $\partial x^*/\partial p$ into substitution and income parts.

Solution. Initial bundle $x^* = y^* = 4$, $u_0 = \sqrt{16} = 4$. Substitution: $h_x = \bar{u} q^{1/2} p^{-1/2}$, $\partial h_x/\partial p = -\bar{u}\sqrt{q}/(2p^{3/2}) = -4/2 = -2$ at $\bar{u} = 4, p = q = 1$. Income: $\partial x^*/\partial M = 1/(2p) = 1/2$, so $-x^* \partial x^*/\partial M = -4(1/2) = -2$. Total $= -2 + (-2) = -4$, which matches $\partial x^*/\partial p = -M/(2p^2) = -4$. Both negative, so x is ordinary, not Giffen.

Problem 6 (applications one-liners)

- (a) In the labor model with non-labor income M and endowment T , what is the reservation wage, and what happens below it?
- (b) In the two-period model with $r = \delta$, what is the relation between c_0 and c_1 , and how do you know whether the consumer saves?

Solution. (a) The reservation wage is the wage at which desired hours first become positive; below it the worker chooses the corner $\ell = T$, $h = 0$, $c = M$. (With $u = \ln c + \ln \ell$ it is $w = M/T$.)

(b) With $r = \delta$, $U'(c_0) = U'(c_1)$ so $c_0 = c_1$ (perfect smoothing). She saves if income is front-loaded ($M_0 > c_0$), i.e. she moves resources to the leaner period. Saving is $s = M_0 - c_0$.

One last thing. If a problem looks unfamiliar, do not panic. Write the budget, write the Lagrangian, take first-order conditions, and read off what the question asks. The steps are always the same. Good luck, you have got this.